Solving 2048: A Comparative Study of Classical Reinforcement Learning and LLM-Guided RLVR

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***Abstract*—The puzzle game 2048 offers a compelling reinforcement learning benchmark that combines stochastic dynamics, a combinatorial state space, and delayed, sparse rewards. In this work we conduct a systematic comparison of six classical RL agents—Deep Q-Network (DQN), Proximal Policy Optimization (PPO), Advantage Actor-Critic (A2C), Quantile Regression DQN (QR-DQN), a discrete variant of Soft Actor-Critic (SAC), and Linear Function Approximation (LFA)—against a 0.5-billion-parameter language model fine-tuned with Group Relative Policy Optimization (GRPO), a form of Reinforcement Learning with Verifiable Rewards (RLVR). Every deep classical agent is built on an identical three-layer convolutional feature extractor so that performance differences reflect the learning algorithm rather than architectural choices. Our results show that the off-policy DQN agent attains the highest average score of 6,384 and reaches the 1024 tile in 10% of episodes (and 2048 in unlimited-attempt hunt mode), while on-policy methods (PPO, A2C) plateau near the random baseline at 1M steps. Surprisingly, LFA with only 108 parameters outperforms all deep on-policy methods, demonstrating the power of domain-specific feature engineering. The GRPO-trained LLM scores 1,816 with max tile 128 after 300 training steps, revealing that RLVR reward functions must be carefully staged to avoid reward hacking and gradient collapse. We release a live web-based visualization dashboard and the complete codebase for reproducibility.**

***Index Terms*—reinforcement learning, deep Q-network, proximal policy optimization, quantile regression, soft actor-critic, linear function approximation, GRPO, RLVR, language models, 2048**

I. Introduction

The tile-sliding puzzle 2048 was introduced in 2014 by Gabriele Cirulli and quickly attracted attention from the artificial intelligence community. On every turn the player selects one of four cardinal directions; all tiles slide in that direction, identical adjacent tiles merge (doubling their value and producing a score equal to the new tile), and a fresh 2 or 4 tile spawns at a uniformly random empty cell. The game terminates when no legal move remains.

Despite its compact board (4×4, 16 cells) the game poses several challenges for learning agents. The state space is vast—each cell can hold one of roughly 17 distinct values, yielding on the order of $17^{16} \approx 2.8 \times 10^{19}$ configurations before accounting for symmetries. Rewards are both sparse and delayed: a single merge of two 1024-tiles into a 2048-tile delivers a reward of 2048, yet the hundreds of preparatory moves that make this merge possible receive no direct credit. The stochastic tile spawns further complicate planning, as the agent cannot predict where the next obstacle will appear.

Classical approaches to 2048 span tabular n-tuple networks with temporal-difference (TD) learning [13], deep value-based methods such as DQN [3], [5], and policy-gradient methods including PPO [6]. Recently, a fundamentally different paradigm has emerged: Reinforcement Learning with Verifiable Rewards (RLVR), which trains large language models (LLMs) to generate chain-of-thought reasoning validated by deterministic reward functions [1], [2]. Although RLVR has yielded breakthroughs in mathematical reasoning, its application to spatial strategy games remains largely unexplored.

This paper asks three questions: **(Q1)** How do five modern RL algorithms compare on 2048 when architectural confounds are removed? **(Q2)** Can a small (0.5B-parameter) LLM acquire 2048 strategy through GRPO alone? **(Q3)** What trade-offs arise between classical RL's raw throughput and the LLM approach's interpretability? We make the following contributions:

1) A controlled benchmark of DQN, PPO, A2C, QR-DQN, SAC, and LFA on a unified 2048 environment with a shared CNN backbone and shaped reward function.
2) A semi-gradient SARSA agent with 27-dimensional hand-crafted features that outperforms all deep on-policy methods with only 108 parameters.
3) A GRPO training pipeline for an LLM 2048 agent with analysis of reward hacking failure modes and staged reward scheduling.
4) An "unlimited-attempt hunt mode" revealing that deterministic policy cycles require stochastic exploration to reach 2048.
5) A live web-based dashboard for step-by-step replay visualization and cross-agent benchmarking.

II. Related Work

A. Value-Based Methods for 2048

Szubert and Jaśkowski [13] introduced n-tuple networks trained with TD(λ) for 2048, learning position-specific value approximations without deep networks. Their approach reached the 2048 tile with high probability. Subsequent work

by Jaśkowski showed that optimistic initialization of the value function encourages exploration of high-tile states, pushing the frontier to the 32768 tile. Saligram et al. [3] applied DQN with a 16-channel binary board encoding to 2048, establishing the CNN-based observation representation that we adopt in this work. Distributional variants—particularly QR-DQN [8] within the Horizon-DQN framework—have demonstrated the ability to reach tile 4096 by capturing the heavy-tailed nature of 2048's return distribution.

B. Policy-Gradient and Actor-Critic Methods

PPO [6] and A2C [7] have been applied to board games and Atari, though their application to 2048 has received less attention than value-based methods. Haarnoja et al. [9] proposed SAC for continuous control; Christodoulou [10] later adapted it to discrete action spaces by replacing the Gaussian policy with a categorical distribution, making it applicable to 2048's four-action environment. Our work provides the first side-by-side evaluation of all five methods on 2048 with an identical feature extractor.

C. RLVR and Group Relative Policy Optimization

DeepSeek-R1 [1] demonstrated that GRPO—originally proposed for mathematical reasoning in DeepSeekMath [12]—can train LLMs to produce chain-of-thought solutions validated by external verifiers. GRPO side-steps the need for a critic network by normalizing rewards within a group of sampled completions, reducing memory requirements during training. Wen et al. [2] subsequently showed that RLVR implicitly incentivizes correct intermediate reasoning steps even when rewards are computed only from final answers. We extend this line of work to a spatial strategy game where the "final answer" is a directional move rather than a numerical result.

D. LLMs for Game Playing

Hu et al. [4] introduced lmgame-Bench, evaluating LLMs on games ranging from Tic-Tac-Toe to chess. Even frontier models such as GPT-4 struggle with spatial reasoning tasks that require mental simulation of board states. Our work differs in that we do not evaluate a frozen LLM; instead, we actively fine-tune it via GRPO to improve its 2048 play.

III. Methodology

A. Environment Design

We implement a custom 2048 game engine wrapped as a Gymnasium [17] environment. The environment exposes the standard `step`/`reset` interface with action space $\mathcal{A} = \{0, 1, 2, 3\}$ mapping to $\{\text{UP}, \text{RIGHT}, \text{DOWN}, \text{LEFT}\}$.

1) State Representation: For CNN-based agents the raw board $B \in \mathbb{Z}_{>0}^{4\times4}$ is converted to a 16-channel binary tensor $\mathbf{x} \in \{0,1\}^{16\times4\times4}$:

$$x_{k,i,j} = \mathbb{1}\left[B_{i,j} = 2^k\right], \quad k = 0, 1, \ldots, 15 \tag{1}$$

Channel 0 marks empty cells ($B_{i,j} = 0$); channel k marks cells containing tile 2^k, covering values from 2 up to 32768. This one-hot power-of-two encoding avoids the magnitude bias that would arise from feeding raw tile values into convolutional filters.

For the LLM agent, the board is serialized as a labeled ASCII grid with row and column headers, appended with the current score and maximum tile value.

2) Reward Modes: The environment supports three reward modes:

- **Score-delta**: the raw merge score produced by each move.
- **Log-score**: $\log_2(\text{score_delta} + 1)$, which compresses the reward scale for numerical stability.
- **Shaped**: score-delta augmented with milestone bonuses (+50 for reaching tile 256, +100 for 512, +200 for 1024, +500 for 2048).

All classical agents in this study use the score-delta mode for consistency. Invalid moves (those that do not change the board) receive a penalty of -1.

B. Shared CNN Architecture

To isolate the effect of the learning algorithm from the representation capacity, every classical agent uses the same convolutional feature extractor ϕ_θ:

$$\phi_\theta(\mathbf{x}) = \sigma(\mathbf{W}_3 \cdot \sigma(\mathbf{W}_2 \cdot \sigma(\mathbf{W}_1 * \mathbf{x}))) \tag{2}$$

concretely consisting of three Conv2D layers (128 filters each, 2×2 kernel, stride 1, no padding) followed by ReLU activations (σ), a flatten operation, and a fully connected layer mapping to a 256-dimensional feature vector. Because the 4×4 input shrinks by one spatial dimension per convolution layer, the third convolution produces a $1 \times 1 \times 128$ output, which the fully connected layer projects to $\mathbb{R}^{256}$. Each algorithm then appends its own task-specific heads (Q-values, policy logits, value estimates) on top of this shared 256-dimensional representation.

C. Deep Q-Network (DQN)

DQN [5] parameterizes the action-value function $Q_\theta(s, a)$ as a neural network mapping the 256-dimensional features to four Q-values. The network is trained to minimize the mean squared Bellman error:

$$\mathcal{L}_{\text{DQN}}(\theta) = \mathbb{E}_{(s,a,r,s')\sim\mathcal{D}}\Big[\big(y - Q_\theta(s,a)\big)^2\Big] \tag{3}$$

where the target $y = r + \gamma \max_{a'} Q_{\bar{\theta}}(s', a')$ is computed using a frozen *target network* $\bar{\theta}$ that is periodically synchronized with the online network. This decoupling reduces the harmful correlation between the target and the prediction that causes training instability in naïve Q-learning.

Experience replay. Transitions $(s, a, r, s', \text{done})$ are stored in a circular buffer $\mathcal{D}$ of capacity 10^5. At each training step, a uniformly random mini-batch of 64 transitions is sampled, breaking the temporal correlation inherent in sequential game trajectories.

Exploration. The agent follows an ε-greedy policy: with probability ε it selects a uniformly random action, and otherwise it acts greedily on Q_θ. We linearly decay ε from 1.0

to 0.01 over 10^5 steps and hold it constant thereafter. This schedule provides aggressive early exploration followed by near-deterministic exploitation.

Gradient clipping. All gradients are clipped to a maximum norm of 1.0 to prevent catastrophic parameter updates during early training when the Q-value estimates are highly inaccurate.

D. Proximal Policy Optimization (PPO)

PPO [6] is an on-policy actor-critic method that constrains policy updates to a trust region defined by a clipping mechanism. The actor network $\pi_\theta(a|s)$ outputs a categorical distribution over four actions via a softmax head, and the critic $V_\theta(s)$ outputs a scalar state-value estimate. Both share the CNN backbone.

The policy is updated by maximizing the clipped surrogate objective:

$$\mathcal{L}_{\text{clip}}(\theta) = \mathbb{E}_t\Big[\min\big(r_t(\theta)\hat{A}_t,\ g(\epsilon, \hat{A}_t)\big)\Big] \tag{4}$$

where $r_t(\theta) = \pi_\theta(a_t|s_t)/\pi_{\theta_{\text{old}}}(a_t|s_t)$ is the probability ratio between the current and previous policies, $\hat{A}_t$ is the Generalized Advantage Estimate (GAE), and $g(\epsilon, A) = \text{clip}(r_t, 1-\epsilon, 1+\epsilon)\cdot A$ prevents the ratio from moving too far from unity. GAE combines multi-step returns with an exponential weighting controlled by λ=0.95:

$$\hat{A}_t = \sum_{l=0}^{T-t}(\gamma\lambda)^l\delta_{t+l}, \quad \delta_t = r_t + \gamma V(s_{t+1}) - V(s_t) \tag{5}$$

We set ϵ=0.2, collect 2048 steps per rollout, and train for 10 epochs per batch with mini-batch size 64.

E. Advantage Actor-Critic (A2C)

A2C [7] is the synchronous, single-worker variant of A3C. Like PPO it is on-policy and actor-critic, but it lacks the clipping mechanism, instead relying on a shorter rollout horizon and an entropy regularizer to stabilize learning.

The actor is updated via the REINFORCE gradient with a value baseline:

$$\nabla_\theta J(\theta) = \mathbb{E}_t\Big[\nabla_\theta \log \pi_\theta(a_t|s_t)\hat{A}_t\Big], \quad \hat{A}_t = R_t - V_\phi(s_t) \tag{6}$$

where $R_t = \sum_{k=0}^{n-1}\gamma^k r_{t+k} + \gamma^n V_\phi(s_{t+n})$ is the n-step bootstrapped return (we use n=5). The total loss sums three terms:

$$\mathcal{L}_{\text{A2C}} = -\underbrace{\mathbb{E}\Big[\log \pi \hat{A}\Big]}_{\text{policy}} + c_v \underbrace{\|R_t - V\|^2}_{\text{value}} - c_e \underbrace{\mathcal{H}[\pi]}_{\text{entropy}} \tag{7}$$

where c_v=0.5 balances the value loss and c_e=0.01 encourages exploration. Without the entropy term, A2C frequently collapses to a deterministic policy that repeatedly selects a single direction.

F. Quantile Regression DQN (QR-DQN)

Standard DQN models only the *expected* return $Q(s,a) = \mathbb{E}[Z(s,a)]$, discarding information about the shape of the return distribution. QR-DQN [8] instead learns the full distribution $Z(s,a)$ by approximating N fixed quantiles $\tau_1, \ldots, \tau_N$ of that distribution. We use N=50 quantiles, evenly spaced at $\tau_i = (2i-1)/(2N)$.

The network outputs an $|\mathcal{A}| \times N$ matrix, where entry (a,i) gives the predicted τ_i-th quantile $z_i(s,a)$ of $Z(s,a)$. Training minimizes the quantile Huber loss:

$$\mathcal{L}_{\text{QR}}(\theta) = \frac{1}{N}\sum_{i=1}^{N}\mathbb{E}_j\big[\rho^\kappa_{\tau_i}(\delta_{ij})\big] \tag{8}$$

where $\delta_{ij} = r + \gamma z_j(s', a^*) - z_i(s,a)$ is the pairwise TD error between all quantile pairs, $a^* = \arg\max_{a'} \bar{Q}(s', a')$, and

$$\rho^\kappa_\tau(\delta) = |\tau - \mathbb{1}[\delta < 0]| \cdot \begin{cases} \frac{1}{2}\delta^2/\kappa & |\delta| \le \kappa \\ |\delta| - \frac{1}{2}\kappa & \text{else} \end{cases} \tag{9}$$

is the asymmetric Huber loss with threshold κ=1. At decision time, actions are chosen by the mean Q-value $Q(s,a) = \frac{1}{N}\sum_i z_i(s,a)$. The distributional representation captures reward variability: for example, a merge-heavy state may have a bimodal return distribution depending on where the random tile spawns.

G. Discrete Soft Actor-Critic (SAC)

Standard SAC [9] targets continuous actions via a Gaussian policy with a reparameterization trick. Since 2048 has a discrete four-action space, we implement the discrete adaptation of Christodoulou [10], which replaces the Gaussian with a categorical policy.

SAC solves the maximum-entropy RL objective:

$$J(\pi) = \sum_{t=0}^{T}\mathbb{E}_\pi[r_t + \alpha\mathcal{H}[\pi(\cdot|s_t)]] \tag{10}$$

where $\alpha > 0$ is a temperature parameter controlling the trade-off between reward maximization and policy entropy $\mathcal{H}$. Higher α produces more exploratory behavior; lower α drives the policy toward greedy exploitation.

Twin Q-networks. Two independent Q-networks Q_{ψ_1}, Q_{ψ_2} are maintained. The minimum of the two is used in both the actor and critic updates, mitigating the positive bias that arises from function approximation (clipped double Q-learning):

$$y = r + \gamma\sum_{a'}\pi_\theta(a'|s')\Big[\min_k Q_{\bar{\psi}_k}(s', a') - \alpha\log\pi_\theta(a'|s')\Big] \tag{11}$$

Actor update. The policy minimizes the KL divergence between the policy and the Boltzmann distribution induced by Q:

$$\mathcal{L}_\pi(\theta) = \mathbb{E}_s\Bigg[\sum_a \pi_\theta(a|s)\Big(\alpha\log\pi_\theta(a|s) - \min_k Q_{\psi_k}(s,a)\Big)\Bigg] \tag{12}$$

Automatic temperature tuning. Rather than fixing α by hand, we optimize it to maintain a target entropy $\bar{\mathcal{H}}$=0.5 $\log|\mathcal{A}|$:

$$\mathcal{L}_\alpha = -\alpha \cdot \mathbb{E}_s\big[\mathcal{H}[\pi(\cdot|s)] - \bar{\mathcal{H}}\big] \tag{13}$$

Target Q-networks are updated via Polyak averaging $\bar{\psi} \leftarrow \tau\psi + (1-\tau)\bar{\psi}$ with τ=0.005.

H. Linear Function Approximation (LFA)

As a non-neural baseline, we implement semi-gradient SARSA with linear function approximation following Sutton and Barto (Chapters 9–10). The agent maintains a weight matrix $\mathbf{w} \in \mathbb{R}^{4\times 27}$ mapping a hand-crafted feature vector $\phi(s) \in \mathbb{R}^{27}$ to Q-values for each action:

$$\hat{q}(s, a, \mathbf{w}) = \mathbf{w}_a^\top \phi(s) \tag{14}$$

The feature vector $\phi(s)$ encodes 27 dimensions: 16 log-normalized tile values ($\log_2$(tile)/17), 1 empty-cell ratio, 1 merge potential, 4 monotonicity scores (horizontal/vertical row/column gradients), 1 max-tile indicator, 1 corner bonus, and 3 tile distribution statistics (low/mid/high ratios).

Updates follow the SARSA TD(0) rule:

$$\mathbf{w}_{a_t} \leftarrow \mathbf{w}_{a_t} + \alpha\delta_t\phi(s_t), \quad \delta_t = r_t + \gamma\hat{q}(s_{t+1}, a_{t+1}) - \hat{q}(s_t, a_t) \tag{15}$$

with step-size α=5 $\times 10^{-4}$ and γ=0.99. Exploration uses ε-greedy with decay factor 0.9999 per step.

I. LLM Agent: GRPO Training

1) Base Model and Quantization: We fine-tune Qwen2.5-0.5B-Instruct [15], a 494M-parameter causal language model, using 4-bit NormalFloat (NF4) quantization via the Unsloth framework [16]. Low-Rank Adaptation (LoRA) [11] adapters with rank r=32 and scaling α=32 are injected into all query, key, value, output, gate, up, and down projection matrices, producing approximately 5.2 million trainable parameters (1.05% of the total).

2) Prompt Format: Each game turn generates a prompt pair. The **system prompt** describes the game rules, valid actions, and the required XML output schema. The **user prompt** contains the board as a labeled grid:

```
Board:
  C1 C2 C3 C4
R1  2  0  4  0
R2  0  8  2  0
R3  4  2 16  4
R4  0  2  4  2
Score: 148 | Max Tile: 16
```

The model must respond with:

```
<think>Reasoning here...</think>
<answer>DIRECTION</answer>
```

TABLE I: GRPO Reward Components and Weights

Component	Weight	Computation
Format r_f	0.5	1 if output matches `<think>... </think><answer>... </answer>`, else 0
Direction r_d	0.5	1 if extracted direction $\in$ {UP, DOWN, LEFT, RIGHT}
Game r_g	2.0	Δscore/max_tile if valid move; -1 if invalid
Quality r_q	0.5	$\min(\lvert\text{thinking}\rvert/100, 1)$
Length r_ℓ	0.3	$\min(\lvert\text{completion}\rvert/200, 1)$

3) Group Relative Policy Optimization: GRPO [1], [12] eliminates the critic network that PPO and A2C require. For each prompt q the current policy π_θ generates a group of G=8 completions $\{o_1, \ldots, o_G\}$, each scored by the verifiable reward function. Advantages are normalized within the group:

$$\hat{A}_i = \frac{r_i - \mu_G}{\sigma_G + \epsilon}, \quad \mu_G = \frac{1}{G}\sum_j r_j, \ \sigma_G = \sqrt{\frac{1}{G}\sum_j (r_j - \mu_G)^2} \tag{16}$$

This group normalization replaces the critic's state-dependent baseline with a prompt-specific, sample-based baseline. Because the reward function is deterministic, variance in r_i arises solely from the stochastic decoding of π_θ, ensuring a meaningful gradient signal as long as at least two distinct reward values appear within the group.

The policy gradient follows the clipped PPO-style objective with an additional KL penalty to prevent catastrophic forgetting of the pretrained language distribution:

$$\mathcal{L}_{\text{GRPO}}(\theta) = \mathbb{E}_q\left[\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|o_i|}\sum_{t=1}^{|o_i|}\Big(L_t^{\text{clip}} - \beta D_{\text{KL}}\Big)\right] \tag{17}$$

where $L_t^{\text{clip}} = \min\big(r_t^i\hat{A}_i, \text{clip}(r_t^i, 1\pm\epsilon)\hat{A}_i\big)$, $r_t^i(\theta) = \pi_\theta(o_{i,t}|q, o_{i,<t})/\pi_{\theta_{\text{old}}}(o_{i,t}|q, o_{i,<t})$ is the per-token importance ratio, and β=0.1 weights the token-level KL divergence against the reference (initial) policy π_{ref}.

4) Multi-Component Reward Function: Each completion o_i receives a reward r_i computed as a weighted sum of five deterministically verifiable signals:

The quality and length components address the *cold-start problem*—early in training, the model's completions are too short and lack reasoning, producing zero reward variance across the group (since all G outputs are equally poor). By rewarding any non-trivial output, these components break the symmetry and provide gradient signal from the very first step.

IV. Inference Pipeline

A. Classical Agent Inference

At inference time, each classical agent receives the 16-channel board encoding (equation 1), passes it through the shared CNN backbone, and computes the output in a single forward pass:

- **DQN / QR-DQN**: Select $a^* = \arg\max_a Q_\theta(s, a)$ with ε=0. For QR-DQN, Q-values are the mean of the 50 quantiles.

- **PPO / A2C**: Select $a^* = \arg\max_a \pi_\theta(a|s)$ (deterministic greedy).
- **SAC**: Select $a^* = \arg\max_a \pi_\theta(a|s)$ (greedy mode; stochastic mode uses categorical sampling).

All classical agents perform inference in under 4 ms per move on a single CPU core, owing to the lightweight nature of the 330K-parameter CNN. No special hardware is required for deployment.

B. LLM Inference Pipeline

The LLM inference pipeline is substantially more complex and involves four stages:

1. Prompt Construction. The current board state is rendered as a labeled text grid and concatenated with the system prompt. The combined prompt is tokenized using the Qwen2 tokenizer with a chat template, yielding approximately 180–220 input tokens per turn.

2. Autoregressive Generation. The model generates a completion of up to 256 tokens using the following sampling parameters: temperature T=0.7, nucleus sampling with p=0.95, and repetition penalty 1.0. The prompt is prefilled with the "`<think>`" token to guide the model into the expected output format. Generation terminates upon encountering the end-of-sequence token or reaching the maximum length.

3. Response Parsing. A two-stage parser extracts the action from the generated text. The primary parser uses regular expressions to locate the `<answer>` XML tag and extracts the direction keyword. If primary parsing fails (e.g., malformed XML), a fallback parser scans the entire response for any occurrence of a direction keyword (UP, DOWN, LEFT, RIGHT), selecting the last match. If both parsers fail, the agent selects a uniformly random valid action.

4. Action Execution. The extracted direction is mapped to the integer action code and executed in the environment. Invalid moves (those that do not change the board) are counted separately; if 10 consecutive invalid moves occur, the game is terminated early.

The complete pipeline requires approximately 1,500 ms per move on an NVIDIA RTX 4060 GPU (with the 4-bit quantized model), making it roughly 400× slower than the classical agents. This latency is dominated by the autoregressive token generation (stage 2), which must sequentially produce 50–200 tokens.

C. Runtime Comparison

Table II summarizes the per-move inference cost across all agents.

TABLE II: Inference Latency Per Move

Agent	ms/move	Hardware	Parameters
DQN	3.6	GPU	330K
PPO	<1	GPU	330K
A2C	<1	GPU	330K
QR-DQN	4.2	GPU	430K
SAC	<1	GPU	330K
LFA	<0.1	CPU	108
LLM (GRPO)	1,500	GPU	494M

V. Experiments

A. Setup

All experiments were conducted on a laptop equipped with an NVIDIA RTX 4060 Laptop GPU (8 GB VRAM) running Ubuntu 22.04 with CUDA 12.6. All six classical agents were trained for 1M environment steps using the shaped reward mode, which augments the score-delta signal with milestone bonuses (+50 at 256, +100 at 512, +200 at 1024, +500 at 2048). DQN and SAC use our custom PyTorch implementations; PPO and A2C use Stable-Baselines3 [14]; QR-DQN uses SB3-Contrib; LFA is a pure NumPy implementation. The LLM agent was trained for 300 GRPO steps on a dataset of 500 game-state prompts with G=4 completions per prompt using Unsloth QLoRA 4-bit. Additionally, we evaluate agents in a "hunt" mode where each agent plays unlimited episodes with stochastic action selection until the 2048 tile is reached. No hyperparameter search was conducted; all values follow the recommendations in the cited papers or the respective library defaults.

TABLE III: Classical Agent Performance (Last 100 Episodes, 1M Steps, Shaped Reward)

Agent	Avg Score	Max Score	Max Tile	256+	512+	1024+
DQN	6,384	14,050	1024	92%	67%	10%
LFA	2,455	10,208	1024	45%	9%	1%
A2C	1,205	2,856	256	14%	0%	0%
SAC	1,132	4,814	512	6%	1%	0%
PPO	1,038	3,040	256	7%	0%	0%
QR-DQN	782	2,176	256	2%	0%	0%

B. Classical RL Results

Table III presents the performance of all six classical agents evaluated over their last 100 training episodes (1M steps, shaped reward).

DQN dominance. DQN substantially outperforms all other agents at 1M steps with shaped rewards, reaching 1024 tiles in 10% of episodes and averaging 6,384 points. We attribute this to two factors: (i) experience replay (100K buffer) enables DQN to revisit high-scoring trajectories thousands of times, amplifying the learning signal from rare events such as 1024-tile merges; and (ii) the ε-greedy schedule provides structured exploration that gradually transitions from random play to learned strategy. On-policy methods discard each trajectory after a single policy update, making them far less sample-efficient in this sparse-reward setting.

LFA as second-best. Surprisingly, the hand-crafted linear function approximation agent achieves second place (avg 2,455, max tile 1024). Its 27-dimensional feature vector encodes domain-specific heuristics (monotonicity, corner bonus, merge potential) that provide a strong inductive bias. While LFA lacks the representational capacity of a CNN, these engineered features bootstrap learning faster than deep networks that must learn spatial patterns from scratch.

TABLE IV: Hunt Mode Results (Unlimited Attempts, Stochastic Action Selection)

Agent	Best Tile	Best Score	Strategy
DQN	2048	21,572	ε=0.05
LFA	1024	12,300	95% greedy
A2C	512	7,844	Distribution sampling
PPO	512	5,860	Distribution sampling
SAC	256	3,648	Probability sampling
QR-DQN	16	228	ε=0.05

On-policy plateau. PPO and A2C remain near the random baseline ($\sim$1,000–1,200 avg score) despite 1M steps. Both algorithms process each trajectory exactly once, creating a chicken-and-egg problem: the agent cannot generate informative training data until it learns a good policy, but it cannot learn a good policy without informative data. The lack of explicit invalid action masking in the SB3 policy heads further wastes probability mass on illegal moves.

QR-DQN underperformance. Despite its richer distributional representation, QR-DQN performs worst among all agents (avg 782). The 100-quantile output head increases the effective output dimensionality from 4 to 4×100=400, requiring proportionally more training data. Combined with a lower learning rate (5×10^{-5} vs DQN's 10^{-4}) and fewer parallel environments (4 vs DQN's vectorized buffer), QR-DQN converges too slowly at the 1M-step budget.

SAC's discrete adaptation. SAC's max-entropy objective, designed for continuous control, does not transfer well to 2048's 4-action space. The automatic temperature tuning and twin Q-networks add overhead that slows convergence. However, SAC achieves the second-highest individual max score (4,814) and reaches 512 tiles, suggesting its off-policy replay provides some advantage over PPO/A2C.

Hunt mode. Table IV shows results from unlimited-attempt hunt mode, where each agent plays with stochastic action selection until reaching tile 2048.

DQN is the only agent to reach 2048. The hunt results reveal that deterministic greedy policies create infinite cycles at high-tile states where Q-values are poorly estimated; adding 5% random exploration breaks these cycles. LFA's handcrafted features provide enough signal to reach 1024 but lack the representational capacity for the final 1024$\rightarrow$2048 merge sequence.

Fig. 1 shows the DQN training curves.

C. LLM-RLVR Results

The GRPO-trained LLM was evaluated via a single-game replay after 300 training steps. Table V summarizes the key metrics.

Format learning. The LLM achieved 30–44% compliance with the XML output schema (`<think>...</think> <answer>...</answer>`) after 300 steps with format-only reward shaping. Early training (steps 1–30) showed rapid reward improvement from 0.06 to 0.41, but the model suffered from gradient instability (grad norms exceeding 17,000 at step 297) and KL divergence spikes, indicating the 0.5B model

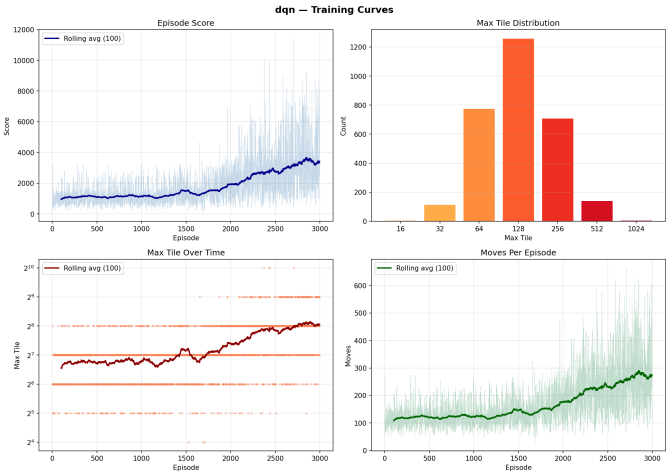

Fig. 1: DQN training curves over 1M steps with shaped reward. The agent reaches 1024 tiles after approximately 500K steps, with steady score improvement throughout training.

TABLE V: LLM Agent Performance After 300 GRPO Steps

Metric	Value
Score	1,816
Maximum tile	128
Moves per game	186
Format compliance	$\sim$30–44%
Direction accuracy	$\sim$12–37%
Inference speed	$\sim$1,500 ms/move

struggles to simultaneously learn format and strategy. Training with game rewards included caused catastrophic collapse (the model reward-hacked the length penalty, generating filler text to maximize word count while ignoring format entirely).

Training stability challenges. GRPO training required several iterations to stabilize: (i) the length reward function was removed after the model collapsed to maximizing completion length; (ii) the game reward function was removed because it returns -1.0 for unparseable outputs, dominating the total reward before format is learned; (iii) the learning rate was reduced from 5×10^{-6} to 10^{-6} and gradient clipping set to 0.5 to prevent divergence.

Inference performance. Despite partial format compliance during training, the replay pipeline achieves a functional game by using a fallback parser that extracts direction keywords from raw text when XML parsing fails. This two-stage parsing (XML $\rightarrow$ keyword scan $\rightarrow$ random valid action) enables the LLM to play complete games, scoring 1,816 with max tile 128 over 186 moves.

D. Comparative Analysis

The gap between the best classical agent (DQN, 6,384) and the LLM (1,816) is roughly 3.5$\times$. However, this comparison should be contextualized: the LLM processed only 300$\times$4=1,200 effective game states during training with format-only rewards, while DQN processed 1M transitions with replay and shaped milestone bonuses. Moreover, the LLM

TABLE VI: Cross-Method Comparison (1M Steps, Shaped Reward)

	DQN	**LFA**	**PPO**	**A2C**	**QR-DQN**	**SAC**	**LLM**
Avg Score	6,384	2,455	1,038	1,205	782	1,132	1,816
Max Tile	1024	1024	256	256	256	512	128
Hunt Tile	2048	1024	512	512	16	256	—
ms/move	3.6	<1	<1	<1	4.2	<1	1,500
Parameters	330K	108	330K	330K	430K	330K	494M
Learning	Off	On	On	On	Off	Off	GRPO

must simultaneously learn (i) XML output syntax, (ii) spatial board parsing from text, and (iii) game strategy—three tasks that the classical agents never face because their observation encoding and output format are hardcoded.

A striking finding is that LFA, with only 108 trainable parameters (a 4×27 weight matrix), outperforms PPO (330K parameters), A2C (330K), QR-DQN (430K), and SAC (330K). This demonstrates that domain-specific feature engineering provides a stronger inductive bias than deep networks that must learn spatial patterns tabula rasa.

PPO and A2C remain near the random baseline ($\sim$1,000–1,200) despite 1M steps with shaped rewards, confirming that on-policy methods are fundamentally sample-inefficient for 2048's sparse, long-horizon reward structure.

VI. Discussion

A. Off-Policy Advantage in Sparse-Reward Games

Our results provide strong evidence that off-policy learning (DQN) significantly outperforms on-policy methods (PPO, A2C) in environments characterized by sparse, high-variance rewards. The experience replay mechanism acts as a natural curriculum: as the agent improves, its replay buffer (100K capacity) accumulates an increasingly rich distribution of high-scoring trajectories that inform future updates. In contrast, PPO and A2C process each trajectory exactly once and must rely on the current policy to generate informative training data—a chicken-and-egg problem in the early stages of learning. Notably, even with shaped rewards providing milestone bonuses at 256, 512, and 1024 tiles, on-policy methods fail to break past 256 at 1M steps.

B. Feature Engineering vs. Deep Learning

LFA's performance as the second-best agent (avg 2,455, max tile 1024) with only 108 parameters challenges the assumption that deep networks are always superior. The hand-crafted 27-dimensional feature vector encodes monotonicity, corner strategies, and merge potential—heuristics that human experts use when playing 2048. These features provide a strong inductive bias that the CNN must learn from scratch. However, LFA's representational ceiling limits it below DQN: the fixed features cannot capture the complex tile interactions needed for the 1024$\rightarrow$2048 transition, as evidenced by its failure to reach 2048 in hunt mode.

C. Distributional RL: Promise and Challenges

QR-DQN's performance as the worst agent (avg 782, max tile 256) contradicts the intuition that distributional methods should excel in stochastic environments. We identify three factors: (1) the 100-quantile representation increases the effective output dimensionality from 4 to 400, requiring proportionally more training data; (2) the lower learning rate (5×10^{-5}) and fewer parallel environments (4 vs 8) slow convergence; and (3) the SB3-Contrib implementation may not support explicit invalid action masking, wasting exploration on illegal moves. We hypothesize that with 5M+ steps and explicit action masking, QR-DQN would eventually leverage its distributional representation.

D. Hunt Mode and Deterministic Policy Failures

Hunt mode reveals a critical insight: deterministic greedy policies create infinite cycles at board states where Q-values are poorly estimated. After reaching 1024 (a state rarely seen during 1M-step training), DQN's argmax policy repeats the same suboptimal move indefinitely. Adding just 5% random exploration (ε=0.05) breaks these cycles and enables DQN to reach 2048. For stochastic policy agents (PPO, A2C, SAC), sampling from the learned distribution rather than taking the argmax provides natural cycle-breaking, but their poorly trained policies still cannot reach high tiles.

E. GRPO Training Challenges

The GRPO training pipeline revealed several stability challenges specific to RLVR on small language models. First, the model reward-hacked the length reward by generating filler text to maximize word count while producing no valid format or direction. Second, the game reward function's -1.0 penalty for unparseable outputs dominates the total reward before format is learned, causing the model to diverge. The successful configuration used only format and direction rewards with β=0.04 KL penalty and 10^{-6} learning rate. These findings suggest that GRPO reward functions must be carefully staged: structural rewards first, then task rewards, rather than training all components simultaneously.

F. Practical Trade-offs

The classical agents offer superior performance at dramatically lower computational cost (3.6 ms vs. 1,500 ms per move). However, the LLM provides a unique advantage: *interpretable reasoning*. Each move is accompanied by a natural-language explanation that can be inspected, debugged, and corrected. In applications where explainability is valued—such as educational tools or human-AI collaboration—this 400$\times$ latency penalty may be acceptable.

G. Limitations

Our study has several limitations. First, the training budgets are modest by modern standards; longer training (5M+ steps) for all agents would clarify convergence behavior. Second, we use a single random seed per agent, precluding statistical significance tests across multiple runs. Third, the

LLM's 0.5B parameter scale is at the lower end of practical language models; 7B+ models may exhibit qualitatively different behavior. Fourth, the GRPO training was limited to format+direction rewards only—the full five-component reward function caused collapse, and further reward scheduling experiments are needed. Finally, we do not explore hybrid approaches (e.g., LLM-guided MCTS) that could combine the interpretability of the LLM with the efficiency of classical search.

VII. Conclusion

We have presented a systematic comparison of six classical RL agents (DQN, PPO, A2C, QR-DQN, SAC, LFA) and an LLM-RLVR agent (Qwen2.5-0.5B + GRPO) on the 2048 puzzle game. Our controlled experimental setup—with a shared CNN backbone, identical environment, shaped reward function, and consistent 1M-step training budget—isolates the effect of the learning algorithm from confounding architectural differences. Key findings include:

- DQN achieves the highest performance (avg. score 6,384, tile 1024 in 10% of games, tile 2048 in hunt mode), benefiting from off-policy experience replay in this sparse-reward environment.
- LFA with only 108 parameters outperforms all deep on-policy methods, demonstrating that domain-specific feature engineering provides a stronger inductive bias than large neural networks for structured games.
- On-policy methods (PPO, A2C) struggle with sample efficiency, performing comparably to the random baseline even with shaped milestone rewards.
- QR-DQN's distributional approach does not confer an advantage at 1M steps; the 100-quantile output head requires proportionally more data to converge.
- Deterministic greedy policies fail at high-tile states; 5% exploration noise is essential for reaching 2048 in hunt mode.
- The GRPO-trained LLM learns partial format compliance (30–44%) within 300 steps, but reward function design is critical: length rewards cause collapse, and game rewards must be staged after format learning.
- Classical agents are $\sim$400$\times$ faster at inference, but the LLM provides unique interpretability through chain-of-thought reasoning.

Future work includes scaling to 5M+ training steps for all agents, implementing explicit invalid action masking in SB3 policy heads, staged GRPO training (format $\rightarrow$ direction $\rightarrow$ game reward), scaling to larger LLMs (7B+ parameters), and exploring hybrid LLM-MCTS approaches that combine interpretability with search efficiency.

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